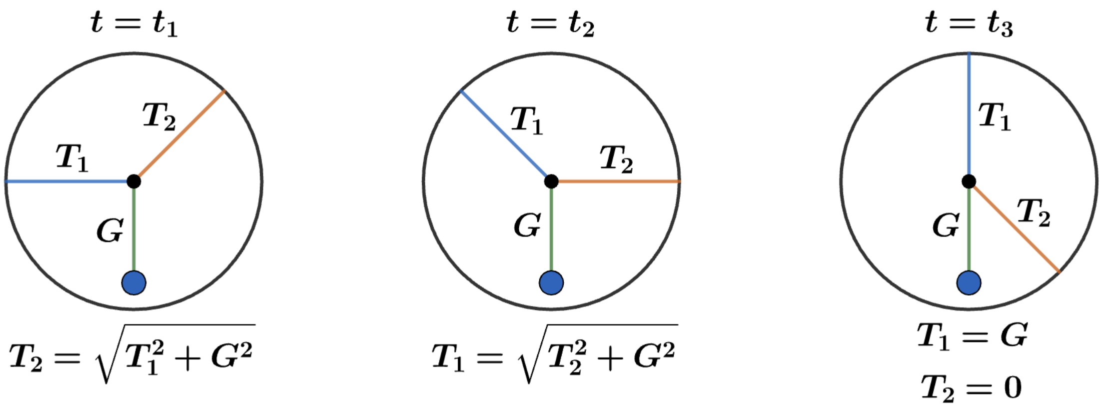
2012 F=ma Exam: Problem 12

Kevin S. Huang



We consider three times $t_1 < t_2 < t_3$ during the rotation. Time t_1 is the original setup where we have

$$T_2 = \sqrt{T_1^2 + G^2} > G$$

by balancing forces. Time t_2 is when the tensions swap,

$$T_1 = \sqrt{T_2^2 + G^2} > G$$

Time t_3 is the final setup where we have

$$T_2 = 0$$

since it can't have a horizontal component $T_{2x} = 0$ which means

$$T_1 = G$$

We now go through the possible choices:

- A) Not correct since comparing t_2 to t_1 shows that T_1 increased.
- B) Not correct since comparing t_3 to t_2 shows that T_1 decreased.
- C) Not correct since comparing t_2 to t_1 shows that T_2 decreased.
- D) Correct since $T_2 = 0$ at time t_3.
- E) Not correct since comparing t_2 to t_1 shows that T_2 is already decreasing in the beginning.

so the answer is $\boxed{\text{D}}$.